Math 54 — Week 2

Vector equations; the matrix equation $A\mathrm{x} = \mathrm{b}$; solution sets (Lay 1.3–1.5)

§1 Facts used from Week 1

(W1) Row operations do not change the solution set of a linear system.

(W2) A system is consistent exactly when an echelon form of its augmented matrix has no row $[\,0 \;\cdots\; 0 \;\; b\,]$ with $b \neq 0$. A consistent system has one solution if there are no free variables, infinitely many otherwise.

(W3) Basic variables correspond to pivot columns; the rest are free. A matrix has at most one pivot per row and per column.

§2 This week

§2.1 Vectors and linear combinations (Lay 1.3)

The question. A linear system asks for numbers $x_1, \ldots, x_n$. Is there a single object, rather than a list of equations, that the system is about?

- A **vector** in $\mathbb{R}^n$ is a column of n real numbers,

$$\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}.$$

  In running text we write it as $(u_1, \ldots, u_n)$. Two vectors are equal when all corresponding entries are equal.
- Sum and scalar multiple are entry by entry: $\mathbf{u} + \mathbf{v}$ has entries $u_i + v_i$, and $c\mathbf{u}$ has entries cu_i. The **zero vector** $\mathbf{0}$ has all entries 0.
- Geometrically (in $\mathbb{R}^2$, $\mathbb{R}^3$): $\mathbf{u} + \mathbf{v}$ is the fourth corner of the parallelogram with sides $\mathbf{u}, \mathbf{v}$; $c\mathbf{u}$ stretches $\mathbf{u}$ by $|c|$, reversing its direction if $c < 0$.
- The usual algebra rules hold: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$, $\mathbf{u} + \mathbf{0} = \mathbf{u}$, $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$, $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$, $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$, $c(d\mathbf{u}) = (cd)\mathbf{u}$, $1\mathbf{u} = \mathbf{u}$. Each follows from the same rule for numbers, applied entry by entry.

Definition. Given vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ in $\mathbb{R}^n$ and scalars $c_1, \ldots, c_p$, the vector $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p$ is a **linear combination** of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ with **weights** $c_1, \ldots, c_p$. The set of all linear combinations is $\mathrm{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$, the subset of $\mathbb{R}^n$ **spanned** by $\mathbf{v}_1, \ldots, \mathbf{v}_p$.

Theorem (A vector equation is a linear system)

The vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$ has the same solution set as the linear system whose augmented matrix is $[\,\mathbf{a}_1 \;\; \mathbf{a}_2 \;\; \cdots \;\; \mathbf{a}_n \;\; \mathbf{b}\,]$. In particular, $\mathbf{b}$ is in $\mathrm{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$ exactly when that system is consistent.

Proof. Two vectors are equal exactly when each entry is equal. Entry i of $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ is $x_1a_{i1} + \cdots + x_na_{in}$, and entry i of $\mathbf{b}$ is b_i. So the vector equation is the list of m equations $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$, whose augmented matrix has the $\mathbf{a}_j$ as columns and $\mathbf{b}$ last. □

Worked example. Let $\mathbf{a}_1 = (1, -2, 3)$, $\mathbf{a}_2 = (0, 1, 2)$, $\mathbf{a}_3 = (2, -5, 4)$ (written as columns). Is $\mathbf{b} = (1, 0, 7)$ in $\mathrm{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$? Is $\mathbf{c} = (4, -9, 14)$?

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ -2 & 1 & -5 & 0 \\ 3 & 2 & 4 & 7 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right], \qquad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ -2 & 1 & -5 & -9 \\ 3 & 2 & 4 & 14 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 4 \end{array}\right].$$

1. For $\mathbf{b}$: consistent, so $\mathbf{b}$ is in the span. Taking the free variable $x_3 = 0$ gives $x_1 = 1$, $x_2 = 2$: $\mathbf{b} = \mathbf{a}_1 + 2\mathbf{a}_2$. Check: $(1, -2, 3) + (0, 2, 4) = (1, 0, 7)$.
2. For $\mathbf{c}$: the last row is $0 = 4$. Not in the span.

Pictures of a span in $\mathbb{R}^3$. $\mathrm{Span}\{\mathbf{v}\}$ for $\mathbf{v} \neq \mathbf{0}$ is the line through $\mathbf{0}$ and $\mathbf{v}$. $\mathrm{Span}\{\mathbf{u}, \mathbf{v}\}$, when neither is a multiple of the other, is the plane through $\mathbf{0}$ containing $\mathbf{u}$ and $\mathbf{v}$. A span always contains $\mathbf{0}$ (all weights 0). In the example above, $\mathbf{a}_3 = 2\mathbf{a}_1 - \mathbf{a}_2$, so the three vectors span only a plane, which contains $\mathbf{b}$ but not $\mathbf{c}$.

§2.2 The matrix equation $A\mathrm{x} = \mathrm{b}$ (Lay 1.4)

Definition. If $A = [\,\mathbf{a}_1 \ \cdots \ \mathbf{a}_n\,]$ is $m \times n$ and $\mathbf{x} \in \mathbb{R}^n$, then $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$. $A\mathbf{x}$ is defined only when the number of columns of A equals the number of entries of $\mathbf{x}$.

- **Row-vector rule**: entry i of $A\mathbf{x}$ is (row i of A)$\cdot\mathbf{x} = a_{i1}x_1 + \cdots + a_{in}x_n$. This is faster by hand. For example $\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix}\begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 4+6-7 \\ 0-15+21 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$.
- The **identity matrix** I_n has 1's on the diagonal and 0 elsewhere; $I_n\mathbf{x} = \mathbf{x}$.

Theorem (Three ways to write one system)

$A\mathbf{x} = \mathbf{b}$, the vector equation $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$, and the linear system with augmented matrix $[\,A \ \ \mathbf{b}\,]$ all have the same solution set. So $A\mathbf{x} = \mathbf{b}$ has a solution exactly when $\mathbf{b}$ is a linear combination of the columns of A.

This is the definition of $A\mathbf{x}$ combined with the theorem in 1.3.

Theorem (When $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$)

For an $m \times n$ matrix A the following are equivalent:

(a) for each $\mathbf{b}$ in $\mathbb{R}^m$, $A\mathbf{x} = \mathbf{b}$ has a solution;
(b) each $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A;
(c) the columns of A span $\mathbb{R}^m$;
(d) A has a pivot position in every row.

The idea

(a), (b), (c) are the same sentence in three languages (previous theorem and the definition of span). The content is (a) $\iff$ (d), which turns a statement about *infinitely many* right-hand sides into a single row reduction of A.

If every row of A has a pivot, then no matter what $\mathbf{b}$ is, the augmented column cannot hold a pivot, because each row's leading entry already sits in A. So no row $[\,0 \cdots 0 \ b\,]$ can appear.

If some row of A has no pivot, the last row of an echelon form U of A is all zeros. Then we can *choose* the right-hand side to make trouble: put 1 in that last position, giving $[\,0 \cdots 0 \ 1\,]$, and run the row operations backward to see which $\mathbf{b}$ that corresponds to. Row operations are reversible, so this works.

Proof. 1. (a), (b), (c) are equivalent by the previous theorem and the definition of Span.

2. (d)⇒(a). Let U be an echelon form of A, reached by some row operations. Apply the same operations to $[\,A\ \ \mathbf{b}\,]$; the result is $[\,U\ \ \mathbf{d}\,]$ for some $\mathbf{d}$. Every row of U has a pivot, so no row of $[\,U\ \ \mathbf{d}\,]$ is $[\,0\cdots0\ b\,]$ with $b\neq0$. By (W2) the system is consistent.
3. not (d) ⇒ not (a). If some row of A has no pivot, the last row of U is zero. Let $\mathbf{d}=(0,\dots,0,1)$; the system $[\,U\ \ \mathbf{d}\,]$ is inconsistent. Apply the reverse row operations to $[\,U\ \ \mathbf{d}\,]$; this gives $[\,A\ \ \mathbf{b}\,]$ for some $\mathbf{b}$, with the same (empty) solution set (W1). So $A\mathbf{x}=\mathbf{b}$ has no solution for this $\mathbf{b}$.

□

- The theorem is about the *coefficient* matrix A, not the augmented matrix. A pivot in every row of $[\,A\ \ \mathbf{b}\,]$ could be a pivot in the last column, which means inconsistency.
- If $m>n$ (more rows than columns), A has at most $n<m$ pivots, so its columns cannot span $\mathbb{R}^m$.

Theorem (Linearity of Ax)
For A $m\times n$, $\mathbf{u},\mathbf{v}\in\mathbb{R}^n$, scalar c: $A(\mathbf{u}+\mathbf{v})=A\mathbf{u}+A\mathbf{v}$ and $A(c\mathbf{u})=c(A\mathbf{u})$.

Proof. $A(\mathbf{u}+\mathbf{v}) = (u_1+v_1)\mathbf{a}_1+\cdots+(u_n+v_n)\mathbf{a}_n = (u_1\mathbf{a}_1+\cdots+u_n\mathbf{a}_n)+(v_1\mathbf{a}_1+\cdots+v_n\mathbf{a}_n)$. Similarly $A(c\mathbf{u})=(cu_1)\mathbf{a}_1+\cdots=c(u_1\mathbf{a}_1+\dots)$. □

§2.3 Solution sets of linear systems (Lay 1.5)

The question. When $A\mathbf{x}=\mathbf{b}$ has infinitely many solutions, what does the set of solutions look like, and how is it related to the solutions of $A\mathbf{x}=\mathbf{0}$?

Homogeneous systems. $A\mathbf{x}=\mathbf{0}$ is **homogeneous**. It always has the **trivial solution** $\mathbf{x}=\mathbf{0}$. A solution with $\mathbf{x}\neq\mathbf{0}$ is **nontrivial**.

- $A\mathbf{x}=\mathbf{0}$ has a nontrivial solution *if and only if* the equation has at least one free variable. (It is always consistent, so by (W2) the solutions are unique exactly when there is no free variable.)

Parametric vector form. Write the general solution as a vector and split it by free variable. For example, take

$$W=\begin{bmatrix}1&3&1\\2&7&4\\1&4&3\end{bmatrix}.$$

Row reduce $[\,W\ \ \mathbf{0}\,]$ and solve:

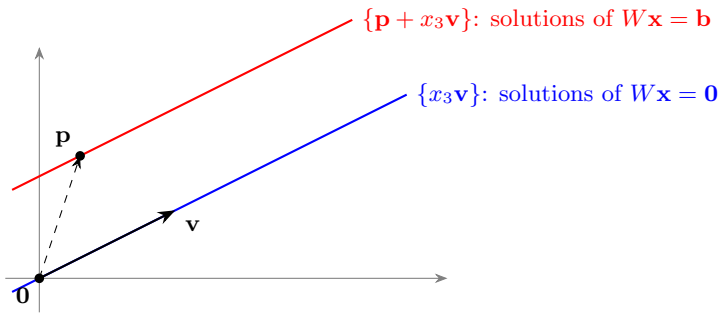
$$\left[\begin{array}{ccc|c}1&3&1&0\\2&7&4&0\\1&4&3&0\end{array}\right]\to\left[\begin{array}{ccc|c}1&0&-5&0\\0&1&2&0\\0&0&0&0\end{array}\right],\quad \mathbf{x}=\begin{bmatrix}5x_3\\-2x_3\\x_3\end{bmatrix}=x_3\begin{bmatrix}5\\-2\\1\end{bmatrix}.$$

The solution set is $\mathrm{Span}\{(5,-2,1)\}$, a line through the origin. In general the solution set of $A\mathbf{x}=\mathbf{0}$ is $\mathrm{Span}\{\mathbf{v}_1,\dots,\mathbf{v}_k\}$ with one vector for each free variable.

Nonhomogeneous systems. Now solve $W\mathbf{x}=(2,5,3)$:

$$\left[\begin{array}{ccc|c}1&3&1&2\\2&7&4&5\\1&4&3&3\end{array}\right]\to\left[\begin{array}{ccc|c}1&0&-5&-1\\0&1&2&1\\0&0&0&0\end{array}\right],\quad \mathbf{x}=\begin{bmatrix}-1+5x_3\\1-2x_3\\x_3\end{bmatrix}=\underbrace{\begin{bmatrix}-1\\1\\0\end{bmatrix}}_{\mathbf{p}}+x_3\begin{bmatrix}5\\-2\\1\end{bmatrix}.$$

The same direction vector appears: the solution set is the line through $\mathbf{p}$ parallel to the solution line of $W\mathbf{x}=\mathbf{0}$.



Schematic: the solutions of $A\mathbf{x} = \mathbf{b}$ are the solutions of $A\mathbf{x} = \mathbf{0}$ shifted by one particular solution $\mathbf{p}$.

Theorem (Solutions of $A\mathbf{x} = \mathbf{b}$ = one solution + solutions of $A\mathbf{x} = \mathbf{0}$)

Suppose $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{p}$ is one solution. Then the solution set of $A\mathbf{x} = \mathbf{b}$ is the set of all $\mathbf{w} = \mathbf{p} + \mathbf{v}_h$, where $\mathbf{v}_h$ is any solution of $A\mathbf{x} = \mathbf{0}$.

The idea

Two solutions of $A\mathbf{x} = \mathbf{b}$ differ by a solution of $A\mathbf{x} = \mathbf{0}$, because A applied to their difference is $\mathbf{b} - \mathbf{b} = \mathbf{0}$. That one subtraction (using linearity) is the whole proof.

Proof. 1. If $A\mathbf{v}_h = \mathbf{0}$, then $A(\mathbf{p} + \mathbf{v}_h) = A\mathbf{p} + A\mathbf{v}_h = \mathbf{b} + \mathbf{0} = \mathbf{b}$. So every $\mathbf{p} + \mathbf{v}_h$ is a solution.
2. If $A\mathbf{w} = \mathbf{b}$, let $\mathbf{v}_h = \mathbf{w} - \mathbf{p}$. Then $A\mathbf{v}_h = A\mathbf{w} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0}$, and $\mathbf{w} = \mathbf{p} + \mathbf{v}_h$. So every solution has this form. □

Writing a solution set in parametric vector form.

1. Row reduce $[\,A \;\; \mathbf{b}\,]$ to reduced echelon form.
2. Express each basic variable in terms of the free variables.
3. Write $\mathbf{x}$ as a vector whose entries depend on the free variables.
4. Split $\mathbf{x}$ into a constant vector plus one vector for each free variable, with that free variable as its weight.

Shortcuts

- **"Is $\mathbf{b}$ a combination of $\mathbf{a}_1, \ldots, \mathbf{a}_n$?"** = "is $[\,\mathbf{a}_1 \cdots \mathbf{a}_n \;\; \mathbf{b}\,]$ consistent?" One row reduction.
- **"Do the columns span $\mathbb{R}^m$?"** = "pivot in every row of A?" Reduce A only; no right-hand side needed.
- **Size check**: an $m \times n$ matrix with $n < m$ never spans $\mathbb{R}^m$; with $n > m$ the homogeneous equation always has a nontrivial solution (a free variable).
- **Homogeneous systems**: never write the zero column during row reduction; it stays zero.
- **Parametric vector form of $A\mathbf{x} = \mathbf{b}$**: the constant vector is a particular solution (free variables 0); the other vectors solve $A\mathbf{x} = \mathbf{0}$. Check by multiplying: $A\mathbf{p} = \mathbf{b}$ and $A\mathbf{v}_i = \mathbf{0}$.
- **One equation in $\mathbb{R}^3$**, $ax_1 + bx_2 + cx_3 = d$, is a plane; with $d = 0$ it passes through $\mathbf{0}$, otherwise it is a parallel plane.

Traps

- $A\mathbf{x}$ needs (columns of A) = (entries of $\mathbf{x}$).
- "Pivot in every row" is about A, not $[\,A \;\; \mathbf{b}\,]$.
- The solution set of $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is *not* a span: it does not contain $\mathbf{0}$.
- $\mathrm{Span}\{\mathbf{u}, \mathbf{v}\}$ is a plane only if neither vector is a multiple of the other; otherwise it is a line (or $\{\mathbf{0}\}$).
- The particular solution $\mathbf{p}$ is not unique; any solution works. Different choices give different-looking but equal sets.

§3 Homework this week

HW1, Section 1.3: 6, 7, 10, 13, 17, 19, 22, 24. Section 1.4: 1, 2, 9, 11, 13, 14, 15, 19, 32, 34. HW2, Section 1.5: 1, 6, 11, 15, 18, 24, 27, 28, 30, 38. (Section 1.7 is in the Week 3 handout.)

§4 Problem set

P1. Write $\mathbf{b} = (5, 1, 12)$ as a linear combination of $\mathbf{a}_1 = (1, 0, 2)$, $\mathbf{a}_2 = (2, 1, 3)$, $\mathbf{a}_3 = (1, -1, 4)$, or show it is impossible.

P2. For which h is $\mathbf{y} = (h, -5, -3)$ in the plane spanned by $\mathbf{v}_1 = (1, 0, -2)$ and $\mathbf{v}_2 = (-3, 1, 8)$?

P3. Do the columns of $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$ span $\mathbb{R}^3$? Do the columns of $B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 4 & 1 \end{bmatrix}$?

P4. Compute $\begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ by the definition (combination of columns) and by the row-vector rule.

P5. Write the solution set of $A\mathbf{x} = \mathbf{0}$ in parametric vector form, for $A = \begin{bmatrix} 1 & 3 & -3 & 7 \\ 0 & 1 & -4 & 5 \\ 2 & 7 & -10 & 19 \end{bmatrix}$.

P6. Write the solution set of $x_1 + 2x_2 - x_3 = 1$, $2x_1 + 5x_2 + x_3 = 4$, $3x_1 + 7x_2 = 5$ in parametric vector form, and describe it geometrically.

P7. True or false, with a reason. (a) If A is 3×2, the columns of A can span $\mathbb{R}^3$. (b) If $A\mathbf{x} = \mathbf{b}$ is consistent for some $\mathbf{b} \neq \mathbf{0}$ and $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then $A\mathbf{x} = \mathbf{b}$ has exactly one solution. (c) If A is 3×5, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

P8. Describe the solution set of the single equation $x_1 + 2x_2 - 3x_3 = 4$ in parametric vector form, and compare it with the solution set of $x_1 + 2x_2 - 3x_3 = 0$.